TRADING BOT SYSTEM SPECIFICATION

Version 1.0 · Bybit V5 Perpetual Execution Engine · Author: @Black · October 2026

1. System Overview & Core Tenets

The algorithmic trading system is an unattended, deterministic execution engine tailored for Bybit USDT Perpetuals (initial target: BTCUSDT) operating under strict Isolated Margin. Built upon a **Hexagonal (Ports & Adapters)** architecture, the core engine enforces written risk models without discretionary emotional bias. The identical business logic executes across Historical Backtests, Bybit Demo Sandboxes, and Live Mainnet.

- **Zero Emotional Drift**: Protective stop-loss and take-profit orders are placed on the exchange engine immediately upon entry.
- **Deterministic Capital Schedule**: Trade risk (R), leverage tiers, and pre-trade friction buffers are mathematical invariants.
- **Fail-Closed Security**: Network errors, invalid order returns, or balance shortfalls trigger immediate position de-risking.

2. System Architecture & Component Model

The system separates core trading logic, orchestration, persistence, and external exchange protocol adapters. Strategies propose signals; the Risk Engine sizes and authorizes; the Exit Policy manages open positions; and Exchange Adapters isolate network protocols.

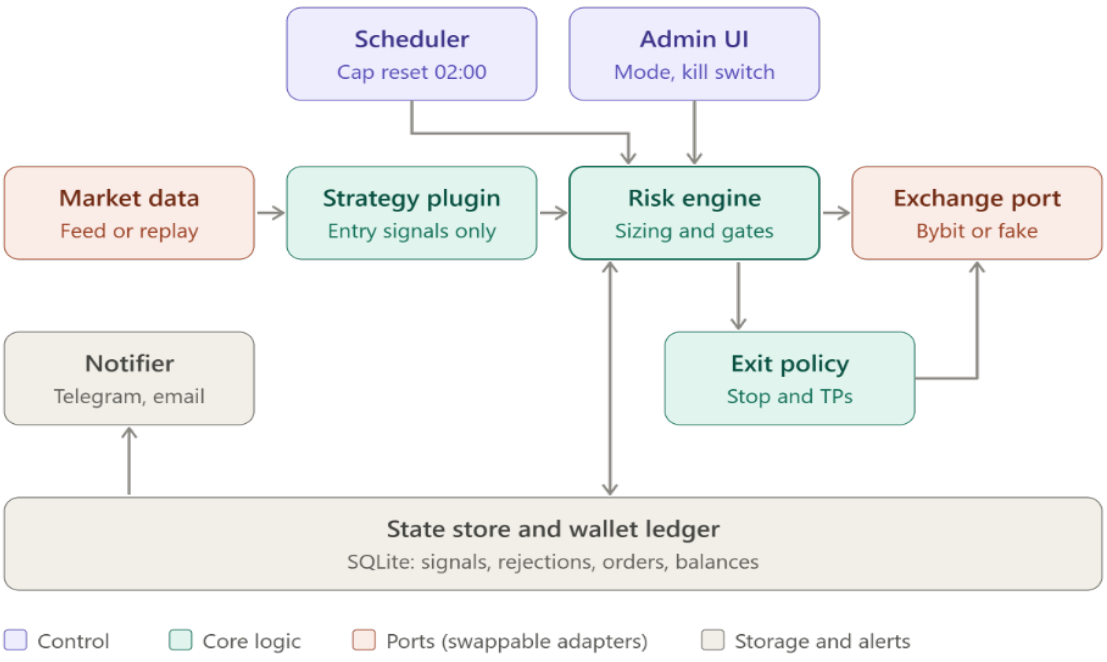


Figure 1: System Decomposition & Inter-Component Signal Pipeline.

3. Operational Use Case Model & System Actors

System actors comprise the Human Administrator (accessing via a secure Web UI) and Autonomous Agents (the Strategy generator and the Midnight Boundary Scheduler).

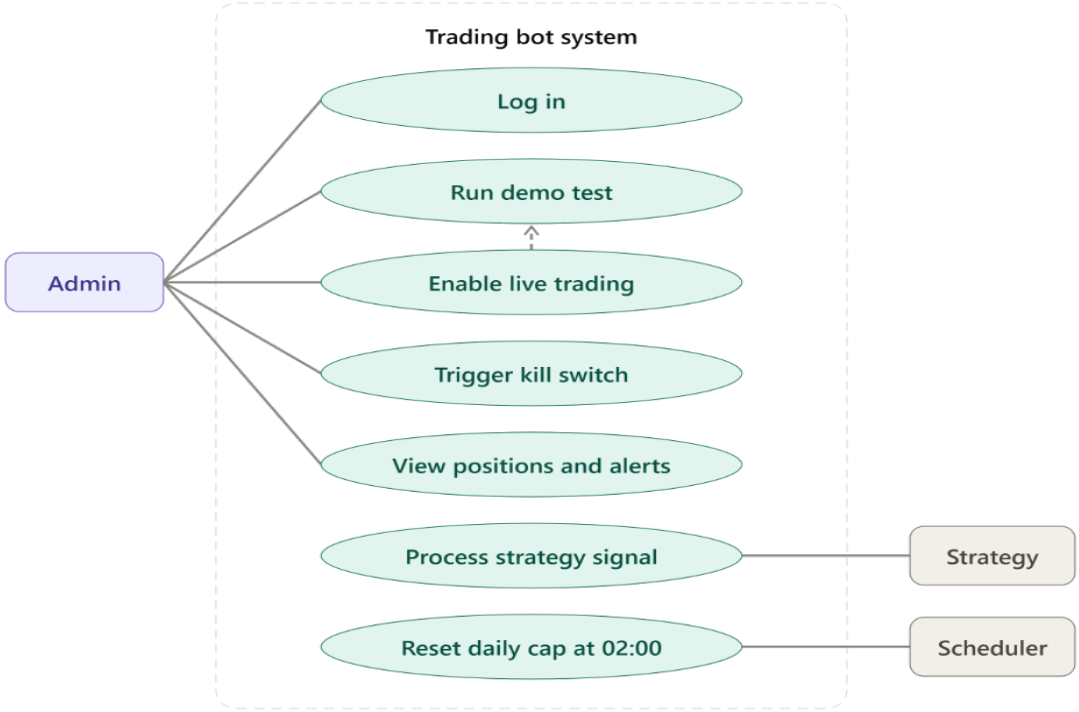


Figure 2: Formal UML Use Case Diagram illustrating administrative and autonomous workflows.

4. Strategy 1: Liquidity Sweep & Momentum Confirmation

Strategy 1 operates across three distinct timeframes: 1-hour macro trend evaluation, 15-minute liquidity sweeps, and 2-minute microstructure confirmation.

1h Trend	15m Trend	Authorized Side	Risk Allocation
Bullish (HH/HL)	Bullish (HH/HL)	Long Only	Full Risk (1.0R)
Bearish (LH/LL)	Bearish (LH/LL)	Short Only	Full Risk (1.0R)
Bullish	Bearish	Short (Counter)	Half Risk (0.5R)
Bearish	Bullish	Long (Counter)	Half Risk (0.5R)
Range / Neutral	Any	Stand Down	No Trade (0.0R)

5. Trade Management & Multi-Stage Exit Policy

Open positions are governed by the Exit Policy engine, which manages limit orders, breakeven adjustments, and momentum giveback protection:

- **Stop Loss**: Placed at Sweep Low minus buffer (10–20 pts; default 15). Exchange reduce-only order.
- **Take Profit 1 (3R)**: Closes ceil(0.70 × Qty / Lot) lots. If that equals total size, exits in full at 3R.
- **Take Profit 2 (5R)**: Closes remaining lots at 5R target.
- **Early Momentum Exit**: Armed once unrealized PnL reaches 1.0R. Closes at market if price gives back ≥ 50% of peak gain prior to 1.5R.
- **Breakeven Shift**: At 1.5R, stop is moved to Entry + Round-Turn Fees + Slippage Buffer.
- **Sweep Invalidation**: If a subsequent 15m candle closes beyond the structural swing point, the trade flattens immediately.

6. Risk Engine & Pre-Trade Gate Hierarchy

Day-start capital (C) at 02:00 Lagos sets the daily risk per trade (R). Setups must pass all 8 gates sequentially; the first failing condition aborts execution and logs to the rejection funnel.

Day-Start Capital (C)	Risk per Trade (R)	Leverage Ceiling	Margin Boundary
Below $10.00	Trading Disabled	None	N/A
$10.00 to $49.99	$1.00 fixed	Up to 100x	≤ 50% Available Equity
$50.00 to $449.99	$1.00 + $0.50 per $50 above $50	Up to 50x	≤ 50% Available Equity
$450.00 to $599.99	$5.00 (Hard Cap)	Up to 50x	≤ 50% Available Equity
$600.00 and above	$5.00 + $1.00 per $100 above $500	Up to 50x	≤ 50% Available Equity

- **Gate 1 (Account Status)**: Equity ≥ $10.00.
- **Gate 2 (Daily Limit)**: Daily trades executed < 2.
- **Gate 3 (Exposure Isolation)**: Zero existing open positions or unfilled orders on the pair.
- **Gate 4 (Directional Bias)**: Multi-timeframe trend allows directional trade.

- **Gate 5 (Exchange Lot Sizing)**: Quantity ≥ 1 lot (0.001 BTC) and Notional ≥ 100 USDT. *Never rounded up*.
- **Gate 6 (Net Reward-to-Risk)**: Net RR = (3d - w) / L ≥ 2.0 (covers round-turn taker/maker fees and slippage).
- **Gate 7 (Margin & Leverage)**: Derived leverage ≤ Tier Ceiling; Initial margin ≤ 50% equity.
- **Gate 8 (Liquidation Distance)**: Distance to estimated liquidation price ≥ 3.0 × Stop distance.

7. Operations Dashboard & Demo-to-Live Qualification Gate

The FastAPI administrative console provides isolated API key management for Bybit Demo and Bybit Live accounts. To protect capital from unvetted configurations, live trading activation is conditionally enforced by code:

- **Independent Sandbox Testing**: Users test strategies on Bybit Demo with zero artificial trade-count restrictions.
- **Mandatory Qualification Gate**: The 'Enable Live Trading' switch remains strictly disabled until the system verifies: **(1)** ≥ **24 hours of continuous demo operation**, and **(2)** ≥ **20 clean demo trades** executed with zero rule violations.
- **Emergency Kill Switch**: A single-click admin trigger immediately flattens open market exposure and disables strategy polling.
- **24-Hour Withdrawal Lock**: Virtual bot wallet requests initiate a mandatory 24-hour security timer prior to releasing funds.

8. Multi-Asset Flexibility Architecture

The system decouples instrument constants into modular PairConfig definitions. Adding secondary pairs (e.g., ETHUSDT, SOLUSDT) requires zero modifications to the core risk engine or state store:

PairConfig(symbol='BTCUSDT', tick_size=0.1, lot_size=0.001, min_notional=100.0, default_buffer=15.0)

9. Infrastructure & Hosting Architecture (Fly.io)

The bot deploys as a lightweight Docker container to Fly.io, configured with an outbound **Static Public IPv4**. This satisfies Bybit's API key security whitelisting requirements and prevents the 90-day key expiration of un-whitelisted keys. State and transaction ledgers are persisted using an encrypted persistent SQLite volume mount (/data).

10. Test-Driven Development (TDD) Verification Plan

Development follows a strict TDD methodology. Core quantitative algorithms must achieve ≥ 95% test coverage prior to exchange integration:

Module	Target Logic	Pass Requirement
risk_schedule.py	Capital tiers & R sizing	Exact match across all capital steps ($9.99 to $600+)
sizing_engine.py	Friction L & Lot rounding	Floor rounding, never round up; reject if < 1 lot
net_rr_gate.py	Taker/Maker fee deduction	Reject setups if Net RR < 2.0
daily_cap.py	2-trade quota & 02:00 reset	3rd trade rejected; correct day boundary tracking
exit_policy.py	Breakeven & Early Exit	Stop moved at 1.5R; 50% peak giveback exit triggered
demo_gate.py	Demo Qualification Gate	Live locked until 24h & 20 clean trades confirmed

***Next Phase Authorization**: Implementation of Phase 1 code will commence upon explicit user confirmation ('Start building').*